ACADEMIC YEAR: 2025-26

Course: Skill base Lab Course: Cloud Computing

Course code: CSL605

Year/Sem: TE/VI

Experiment No.: 06
Aim: To study and Implement Database as a Service on SQL/NOSQL databases using MongoDB.
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Roll Number: 40
Date of Performance: 16/02/2026
Date of Submission: 18/02/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	2
Understanding	4-5	2-3	2
Journal work and timely submission.	8-10	5-8	4

Checked by

Name of Faculty : Dr. Tatwadarshi Nagarhalli

Signature :

Date :

Experiment No. 6

Aim: To study and Implement Database as a Service on SQL/NOSQL databases like AWS RDS, AZURE SQL/ MongoDB Lab/ Firebase.

Objective:

The objective of this experiment is to study the concept of Database as a Service (DBaaS) in cloud computing and to implement a NoSQL database using MongoDB. It also aims to deploy the database on an Amazon EC2 instance in the Amazon Web Services cloud environment and perform basic database operations such as creating databases, collections, and inserting documents.

Theory:

NoSQL Cloud Database Services offer cloud-based solutions optimized for scalability, performance, and cost efficiency. They are designed to manage diverse data types while supporting rapid growth and high availability.

- Handle structured, semi-structured, and unstructured data efficiently
- Scale horizontally to manage large and growing data volumes
- Ideal for high-performance, distributed, and cloud-native applications

Advantages of Cloud-Based NoSQL Databases

1. Scalability: Easily scale horizontally by adding servers as data grows.

2. High Availability: Ensures reliable uptime and consistent performance.

3. Cost Efficiency: Reduces expenses by eliminating on-premise hardware and infrastructure.

4. High Performance: Delivers low latency and fast data access.

5. Flexibility: Supports structured, semi-structured, and unstructured data.

Some popular NoSQL Cloud Database Services include:

1. Amazon DynamoDB: Fully managed NoSQL database with fast, predictable performance and seamless scalability.

2. Google Cloud Datastore: Managed document-based NoSQL database with automatic scaling and low latency.

3. Microsoft Azure Cosmos DB: Globally distributed, multi-model database offering high availability and low latency.

4. MongoDB Atlas: Fully managed MongoDB service with automated backups, security, and easy scaling.

A NoSQL Cloud Database Services is a database that provides a means that helps to store and retrieve data is a NoSQL database. The term non-relational/non-SQL is initially referred to as NoSQL. Big data and web applications are some of its application that uses NoSQL databases in real-time and their utilization to boost with time. Sometimes, NoSQL databases are referred to as 'Not only SQL' because query languages related to SQL are also supported. Better command over accessibility, less difficult level scaling to bunches of devices, and design simplicity are included in a NoSQL database.

Types of NoSQL databases:

Column-based

Key-value store

Graph databases

Document-Base

d

Types of NoSQL databases

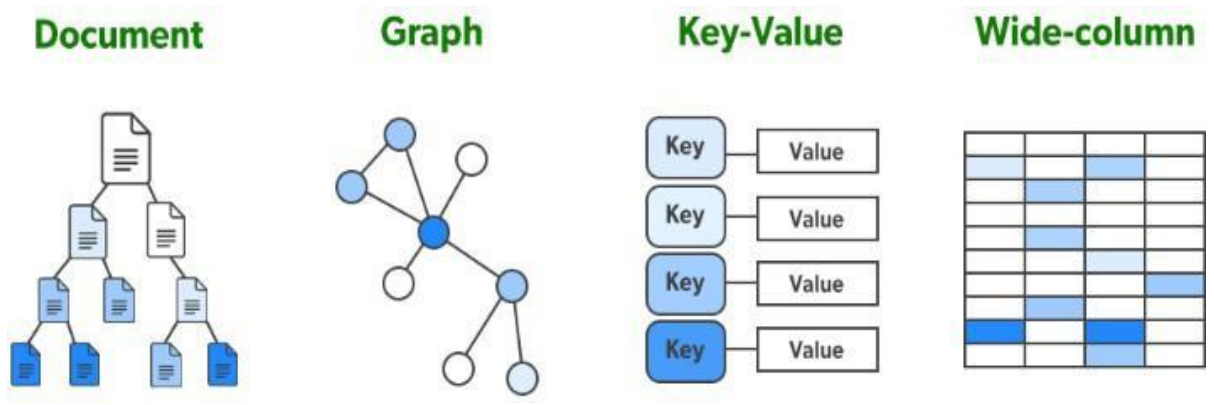


Figure 1: Types of NoSQL databases

Choosing NoSQL:

Google, Facebook, Amazon, and Linkedln are a few of the top internet companies that

originally use the NoSQL database for overcoming the downside of the Relational Database Management System(RDBMS). Relational Database Management System isn't dependably the best answer for all circumstances since data handling necessities develop dramatically. A dynamic method and a better cloud-friendly environment are provided by NoSQL for processing unstructured data.

Difference between NoSQL Databases and SQL Databases:

- NoSQL database schema is flexible while SQL database schema is rigid.
- In NoSQL databases, queries are more immediate than SQL databases.
- NoSQL databases are non-relational while SQL databases are relational.
- NoSQL Examples -HBase, Bigtable.
- SQL Examples- Sybase, Access, Oracle, and MySQL.

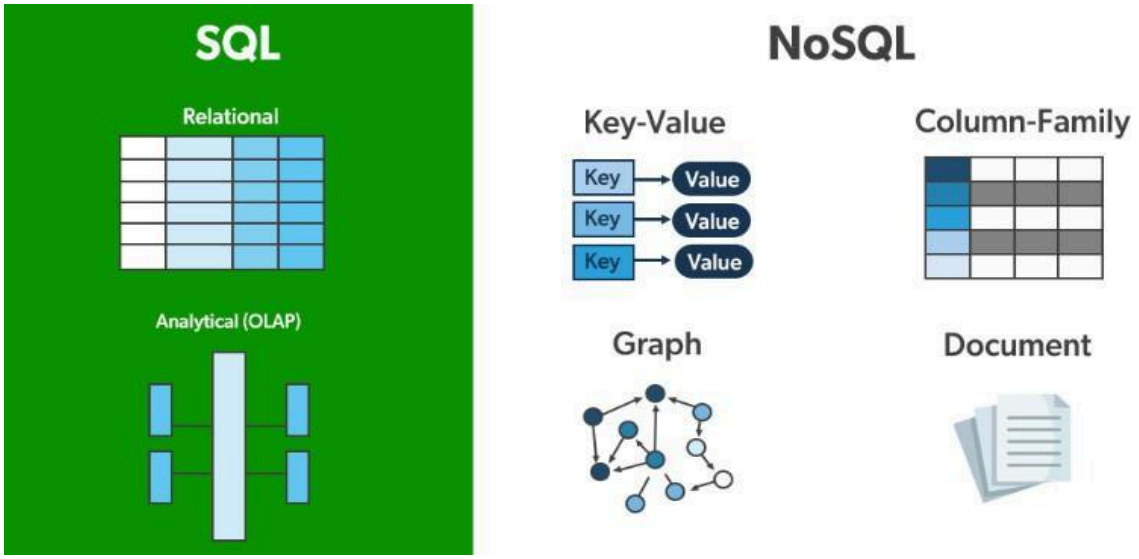


Figure 2: Difference between SQL and NoSQL Databases **Steps to Implement DBaas in MongoDB Atlas (DBaaS) Step 1: Login to AWS**

1. Open a browser.
2. Go to **AWS Management Console**.
3. Sign in to your **AWS account**.

Step 2: Launch EC2 Instance

1. In the search bar, open **Amazon EC2**.
2. Click **Launch Instance**.
3. Enter **Instance Name** (Example: MongoDBServer).
4. Choose **Ubuntu Server** as the operating system.

5. Select **t2.micro** instance type (Free Tier eligible).
6. Create a **Key Pair (.pem file)** and download it.
7. In **Network Settings**,
   allow: o SSH
   (Port 22)
8. Click **Launch Instance**.

Step 3: Connect to EC2 from Windows

1. Open the EC2 dashboard.
2. Select your instance.
3. Click **Connect**.
4. Choose **EC2 Instance Connect (browser-based terminal)**.

(This is easiest for Windows users because no extra software is required.)

Step 4: Update the Server
Run:
sudo apt update

Step 5: Install MongoDB
Run:
sudo apt install mongodb -y

Step 6: Start MongoDB Service
sudo systemctl start mongodb
Enable auto start:
sudo systemctl enable mongodb

Step 7: Open MongoDB Shell
mongosh

Step 8: Create Database
use collegeDB

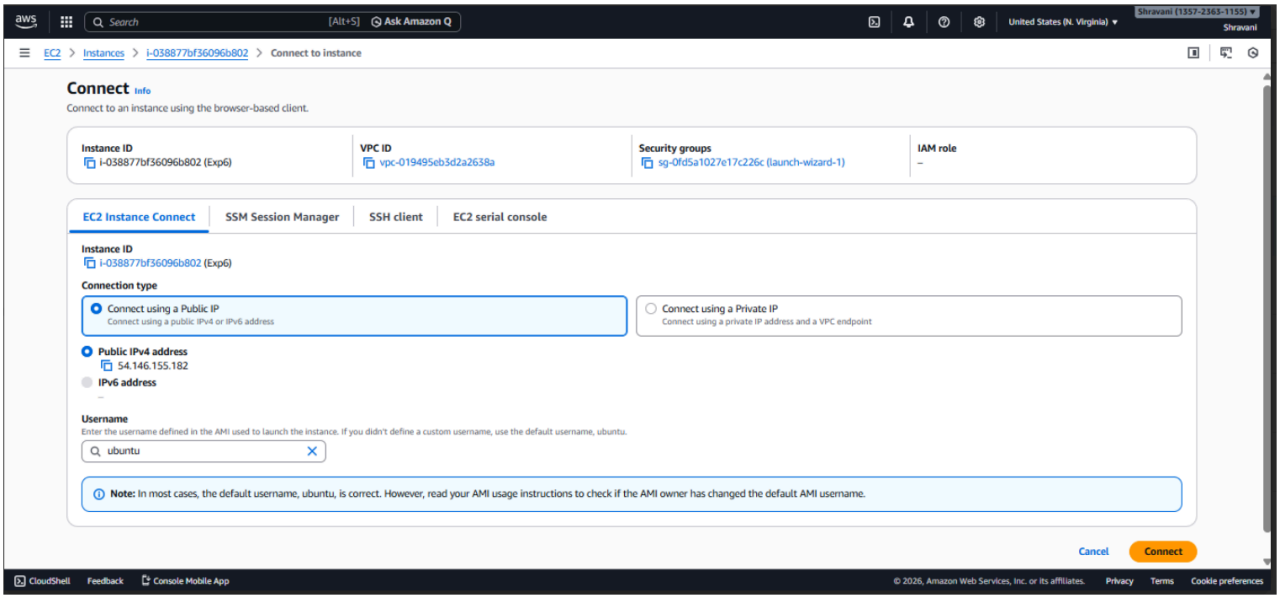
Step 9: Create Collection
db.createCollection("students")

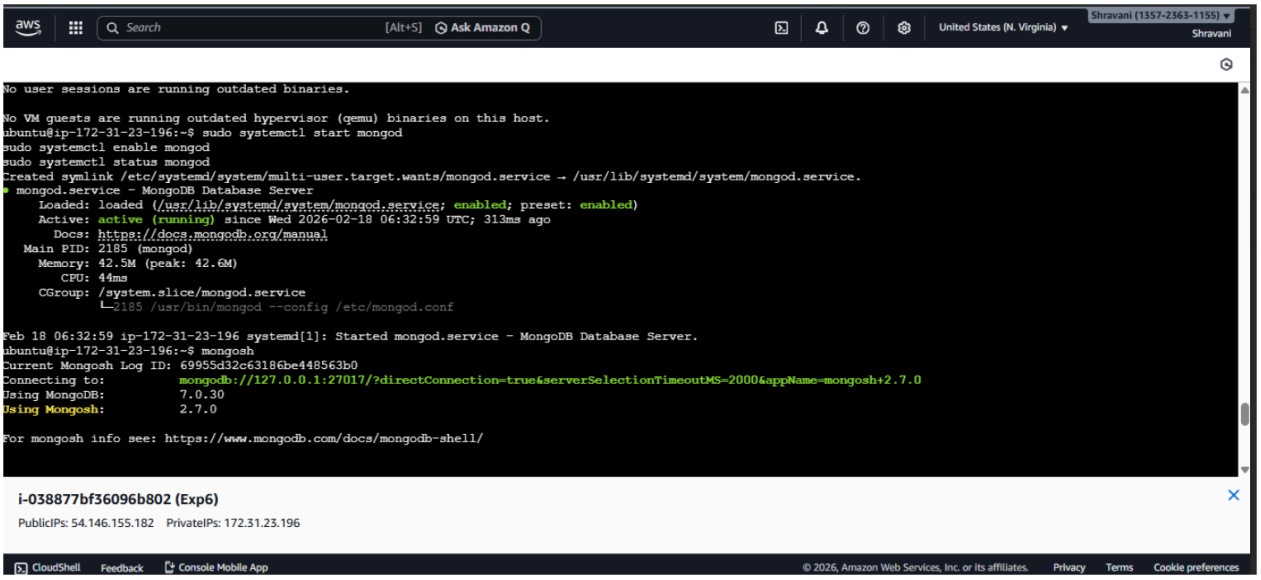
Step 10: Insert Document

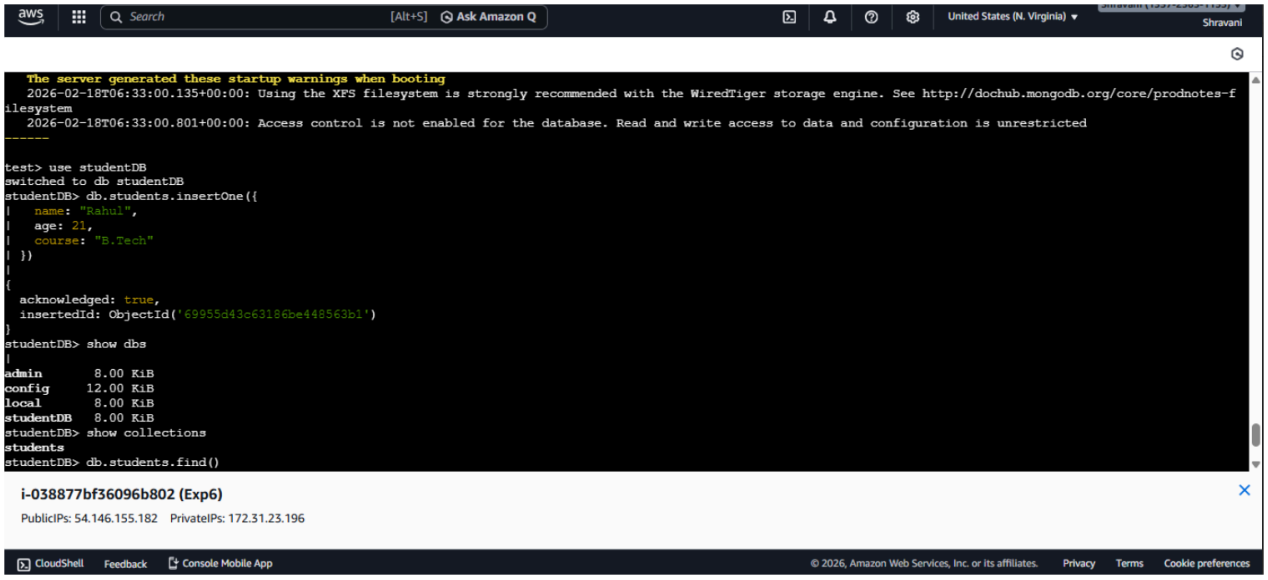
```
db.students.insertOne({
name:"Riya",
roll_no:101,
course:"Computer Engineering"
})
```

Step 11: Display Data
db.students.find()

Output/Observation:







Conclusion:

SQL databases use a structured, table-based format with predefined schemas and support complex queries using SQL, making them suitable for applications requiring strong consistency and relationships. In contrast, NoSQL databases are schema-less, flexible, and designed for scalability, handling unstructured or semi-structured data efficiently. MongoDB plays a key role as a NoSQL database by storing data in JSON-like documents, allowing dynamic schemas, high scalability, and fast performance, making it ideal for modern applications like real-time analytics, big data, and web development.